```
D:\conda\envs\AutoML\python.exe D:\programy_studia\AutoML\projekt_2\AutoML-system-binary-classification\sample_data_test.py
   id  gender  SeniorCitizen Partner Dependents tenure PhoneService  \
0   0    Male              0     Yes        Yes     29          Yes
1   1    Male              0     Yes        Yes     58          Yes
2   2    Male              0     Yes         No     58          Yes
3   3  Female              0      No         No      1          Yes
4   4  Female              0      No         No      1          Yes

  MultipleLines InternetService OnlineSecurity OnlineBackup DeviceProtection  \
0            No             DSL            Yes            No              Yes
1            No             DSL            Yes           Yes               No
2           Yes     Fiber optic             No           Yes               No
3            No     Fiber optic             No            No               No
4            No     Fiber optic             No            No               No

  TechSupport StreamingTV StreamingMovies Contract PaperlessBilling  \
0         Yes          No               No        One year              Yes
1         Yes         Yes               No        Two year               No
2          No         Yes              Yes  Month-to-month              Yes
3          No          No               No  Month-to-month              Yes
4          No          No               No  Month-to-month              Yes
```

```

              PaymentMethod  MonthlyCharges  TotalCharges Churn
0             Mailed check           60.10       1653.85    No
1  Credit card (automatic)           69.50       3778.20    No
2         Electronic check          100.40       5841.35    No
3         Electronic check           69.70         70.70   Yes
4         Electronic check           70.45         70.45   Yes
Begin preprocessing...
--- KMeans: Wytrenowano 15 klastrów ---
   -> Wybieranie top 6 cech do interakcji (ExtraTrees)...
   Wybrane kolumny do interakcji: ['MonthlyCharges', 'TotalCharges', 'FREQ_tenure', 'Dist_Cluster_8', 'Dist_Cluster_3', 'Dist_Cluster_13']
--- Współliniowość: Usunięto 48 z 129 kolumn (korelacja > 0.95) ---

--- Selekcja Cech (Hybrid) ---
-> Etap 1: Wstępne usuwanie szumu (L1 + ExtraTrees)...
   -> Ocena Liniowa (L1 Logistic Regression)...
   -> Ocena Nieliniowa (ExtraTreesClassifier)...
   L1 wybrało: 41, Trees wybrało: 55
   Wspólna decyzja: Zachowano 61 z 81 cech (usunięto 20 najsłabszych).
-> Etap 2: Liczba cech (61) <= 150. Uruchamiam SFS Backward.
-> SFS zakończony. Wybrano 25 najlepszych cech.
--- Selekcja Cech Zakończona: Tryb = sfs ---

Ostateczne kolumny:['SQR_MonthlyCharges', 'MUL_TotalCharges_x_Dist_Cluster_3', 'ADD_Dist_Cluster_3_Dist_Cluster_13', 'TotalCharges', 'Dist_Cluster_1', 'DIV_Dist_Cluster_13_by_Dist_Cluster_8', '
```

```
Dist_Cluster_0', 'LOG_TotalCharges', '
MUL_MonthlyCharges_x_TotalCharges', 'MonthlyCharges
', 'ADD_MonthlyCharges_TotalCharges', '
LOG_MonthlyCharges', 'gender', 'Dependents', '
BIN_Dist_Cluster_13', 'MultipleLines', 'Contract', '
StreamingTV', 'DeviceProtection', 'OnlineSecurity', '
InternetService', 'TechSupport', 'PhoneService', '
PaperlessBilling', 'PaymentMethod']

Preprocessing done.
--- Stage 1: Screening 28 models ---
log_reg_tabarena_1 → BA = 0.7670
log_reg_tabarena_2 → BA = 0.7674
log_reg_tabarena_3 → BA = 0.7670
log_reg_tabarena_4 → BA = 0.7669
log_reg_tabarena_4 → BA = 0.7673
knn_tabarena → BA = 0.7706
randomforest_tabarena_1 → BA = 0.7691
randomforest_tabarena_2 → BA = 0.7645
randomforest_tabarena_3 → BA = 0.7618
extraTree_tabarena_3 → BA = 0.7690
xgboost_tabarena_1 → BA = 0.7774
xgboost_tabarena_2 → BA = 0.7349
xgboost_FLAML_1 → BA = 0.7771
xgboost_FLAML_2 → BA = 0.7799
xgboost_FLAML_3 → BA = 0.7653
xgboost_FLAML_4 → BA = 0.7228
catboost_tabarena_1 → BA = 0.7653
catboost_tabarena_2 → BA = 0.7741
catboost_tabarena_3 → BA = 0.7780
catboost_memento → BA = 0.7456
lightgbm_tabarena_1 → BA = 0.7775
lightgbm_tabarena_2 → BA = 0.7355
lightgbm_FLAML_1 → BA = 0.7671
lightgbm_FLAML_2 → BA = 0.7736
lightgbm_FLAML_3 → BA = 0.7792
lightgbm_memento → BA = 0.7732
NeuralNetTorch_tabarena_1 → BA = 0.7996
NeuralNetTorch_tabarena_2 → BA = 0.7189

--- Stage 2: Stacking Ensembles ---
```

```
Stacking Top 3 Best → 0.9100
Stacking Top 3 Unique → 0.9100 (Models: 3)

--- Stage 3: Fast Heuristics (Top 5) ---
Ensemble: Mean Top 5 → 0.9113
Ensemble: Rank Avg Top 5 → 0.9114
Ensemble: Champ+Rebel → 0.9086

==============================
WINNER: Ensemble: Rank Avg Top 5
Balanced Accuracy: 0.9114
==============================
Final fitting of Ensemble: Rank Avg Top 5...
Leaderboard:
================ Leaderboard ================
                          Model Name  \
0          Ensemble: Rank Avg Top 5
1              Ensemble: Mean Top 5
2   Ensemble: Stacking Top 3 Unique
3     Ensemble: Stacking Top 3 Best
4             Ensemble: Champ+Rebel
5         NeuralNetTorch_tabarena_1
6                  xgboost_FLAML_2
7                 lightgbm_FLAML_3
8              catboost_tabarena_3
9              lightgbm_tabarena_1
10              xgboost_tabarena_1
11                 xgboost_FLAML_1
12             catboost_tabarena_2
13                lightgbm_FLAML_2
14                lightgbm_memento
15                    knn_tabarena
16          randomforest_tabarena_1
17             extraTree_tabarena_3
18              log_reg_tabarena_2
19              log_reg_tabarena_4
20                lightgbm_FLAML_1
21              log_reg_tabarena_1
22              log_reg_tabarena_3
23              log_reg_tabarena_4
24                 xgboost_FLAML_3
```

```
25              catboost_tabarena_1
26          randomforest_tabarena_2
27          randomforest_tabarena_3
28                 catboost_memento
29              lightgbm_tabarena_2
30               xgboost_tabarena_2
31                  xgboost_FLAML_4
32        NeuralNetTorch_tabarena_2

                                     Model Class
    Metric Score  \
0                                      Heuristic
         0.911357
1                                      Heuristic
         0.911291
2                                       Ensemble
         0.909958
3                                       Ensemble
         0.909958
4                                      Heuristic
         0.908591
5   wrappers.TorchNNClassifier.TorchNNClassifier
         0.799648
6                         xgboost.XGBClassifier
         0.779852
7                       lightgbm.LGBMClassifier
         0.779246
8                   catboost.CatBoostClassifier
         0.778021
9                       lightgbm.LGBMClassifier
         0.777485
10                        xgboost.XGBClassifier
         0.777365
11                        xgboost.XGBClassifier
         0.777146
12                  catboost.CatBoostClassifier
         0.774093
13                      lightgbm.LGBMClassifier
         0.773632
14                      lightgbm.LGBMClassifier
         0.773161
```

```
145 15        sklearn.neighbors.KNeighborsClassifier
          0.770553
146 16       sklearn.ensemble.RandomForestClassifier
          0.769051
147 17         sklearn.ensemble.ExtraTreesClassifier
          0.768965
148 18       sklearn.linear_model.LogisticRegression
          0.767381
149 19       sklearn.linear_model.LogisticRegression
          0.767328
150 20                      lightgbm.LGBMClassifier
          0.767116
151 21       sklearn.linear_model.LogisticRegression
          0.767037
152 22       sklearn.linear_model.LogisticRegression
          0.766953
153 23       sklearn.linear_model.LogisticRegression
          0.766866
154 24                        xgboost.XGBClassifier
          0.765349
155 25                   catboost.CatBoostClassifier
          0.765323
156 26       sklearn.ensemble.RandomForestClassifier
          0.764499
157 27       sklearn.ensemble.RandomForestClassifier
          0.761785
158 28                   catboost.CatBoostClassifier
          0.745556
159 29                      lightgbm.LGBMClassifier
          0.735472
160 30                        xgboost.XGBClassifier
          0.734862
161 31                        xgboost.XGBClassifier
          0.722846
162 32  wrappers.TorchNNClassifier.TorchNNClassifier
          0.718918
163
164
    Wrapper  \
165 0   <MiniAutoML.HeuristicEnsemble object at 0x0000
    ...
```

```
1   <MiniAutoML.HeuristicEnsemble object at 0x0000
...
2   <MiniAutoML.StackingEnsemble object at 0x00000
...
3   <MiniAutoML.StackingEnsemble object at 0x00000
...
4   <MiniAutoML.HeuristicEnsemble object at 0x0000
...
5   <wrappers.wrapper_model.ModelWrapper object at
...
6   <wrappers.wrapper_model.ModelWrapper object at
...
7   <wrappers.wrapper_model.ModelWrapper object at
...
8   <wrappers.wrapper_model.ModelWrapper object at
...
9   <wrappers.wrapper_model.ModelWrapper object at
...
10  <wrappers.wrapper_model.ModelWrapper object at
...
11  <wrappers.wrapper_model.ModelWrapper object at
...
12  <wrappers.wrapper_model.ModelWrapper object at
...
13  <wrappers.wrapper_model.ModelWrapper object at
...
14  <wrappers.wrapper_model.ModelWrapper object at
...
15  <wrappers.wrapper_model.ModelWrapper object at
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16  <wrappers.wrapper_model.ModelWrapper object at
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17  <wrappers.wrapper_model.ModelWrapper object at
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18  <wrappers.wrapper_model.ModelWrapper object at
...
19  <wrappers.wrapper_model.ModelWrapper object at
...
20  <wrappers.wrapper_model.ModelWrapper object at
...
21  <wrappers.wrapper_model.ModelWrapper object at
```

```
...
22  <wrappers.wrapper_model.ModelWrapper object at
...
23  <wrappers.wrapper_model.ModelWrapper object at
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24  <wrappers.wrapper_model.ModelWrapper object at
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25  <wrappers.wrapper_model.ModelWrapper object at
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26  <wrappers.wrapper_model.ModelWrapper object at
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28  <wrappers.wrapper_model.ModelWrapper object at
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...
30  <wrappers.wrapper_model.ModelWrapper object at
...
31  <wrappers.wrapper_model.ModelWrapper object at
...
32  <wrappers.wrapper_model.ModelWrapper object at
...


Config  \
0
                                                                {}

1
                                                                {}

2
                                                                {}

3
                                                                {}

4
                                                                {}
```

```
5   {'name': 'NeuralNetTorch_tabarena_1', 'class':...
6   {'name': 'xgboost_FLAML_2', 'class': 'xgboost....
7   {'name': 'lightgbm_FLAML_3', 'class': 'lightgb...
8   {'name': 'catboost_tabarena_3', 'class': 'catb...
9   {'name': 'lightgbm_tabarena_1', 'class': 'ligh...
10  {'name': 'xgboost_tabarena_1', 'class': 'xgboo...
11  {'name': 'xgboost_FLAML_1', 'class': 'xgboost....
12  {'name': 'catboost_tabarena_2', 'class': 'catb...
13  {'name': 'lightgbm_FLAML_2', 'class': 'lightgb...
14  {'name': 'lightgbm_memento', 'class': 'lightgb...
15  {'name': 'knn_tabarena', 'class': 'sklearn.nei...
16  {'name': 'randomforest_tabarena_1', 'class': '...
17  {'name': 'extraTree_tabarena_3', 'class': 'skl...
18  {'name': 'log_reg_tabarena_2', 'class': 'sklea...
19  {'name': 'log_reg_tabarena_4', 'class': 'sklea...
20  {'name': 'lightgbm_FLAML_1', 'class': 'lightgb...
21  {'name': 'log_reg_tabarena_1', 'class': 'sklea...
22  {'name': 'log_reg_tabarena_3', 'class': 'sklea...
23  {'name': 'log_reg_tabarena_4', 'class': 'sklea...
24  {'name': 'xgboost_FLAML_3', 'class': 'xgboost....
25  {'name': 'catboost_tabarena_1', 'class': 'catb
```

```
...
26  {'name': 'randomforest_tabarena_2', 'class': '...
27  {'name': 'randomforest_tabarena_3', 'class': '...
28  {'name': 'catboost_memento', 'class': 'catboos...
29  {'name': 'lightgbm_tabarena_2', 'class': 'ligh...
30  {'name': 'xgboost_tabarena_2', 'class': 'xgboo...
31  {'name': 'xgboost_FLAML_4', 'class': 'xgboost....
32  {'name': 'NeuralNetTorch_tabarena_2', 'class':...


Params
0                                        Rank Averaging
1                                           Simple Mean
2   Unique Models: ['NeuralNetTorch_tabarena_1', '...
3                                   Stacking Top 3 Best
4                                Champ & #5 (Corr 0.94)
5   {'activation': 'relu', 'batch_size': 64, 'drop...
6   {'objective': 'binary:logistic', 'base_score':...
7   {'boosting_type': 'gbdt', 'class_weight': None...
8   {'learning_rate': 0.01854744499982, 'depth': 5...
9   {'boosting_type': 'gbdt', 'class_weight': None...
10  {'objective': 'binary:logistic', 'base_score':...
11  {'objective': 'binary:logistic', 'base_score
```

```
':...
12  {'learning_rate': 0.055078095725390006, 'depth
...
13  {'boosting_type': 'gbdt', 'class_weight': None
...
14  {'boosting_type': 'gbdt', 'class_weight': None
...
15  {'algorithm': 'auto', 'leaf_size': 30, 'metric
...
16  {'bootstrap': True, 'ccp_alpha': 0.0, 'class_w
...
17  {'bootstrap': False, 'ccp_alpha': 0.0, 'class_
...
18  {'C': 857.9598230609945, 'class_weight': None
,...
19  {'C': 1.0, 'class_weight': None, 'dual': False
...
20  {'boosting_type': 'gbdt', 'class_weight': None
...
21  {'C': 758.6397627599249, 'class_weight': None
,...
22  {'C': 0.553292480550028, 'class_weight': None
,...
23  {'C': 0.6309573444801311, 'class_weight': None
...
24  {'objective': 'binary:logistic', 'base_score
':...
25  {'learning_rate': 0.030421683021409, 'depth
': ...
26  {'bootstrap': True, 'ccp_alpha': 0.0, 'class_w
...
27  {'bootstrap': True, 'ccp_alpha': 0.0, 'class_w
...
28  {'iterations': 278, 'learning_rate': 0.7293529
...
29  {'boosting_type': 'gbdt', 'class_weight': None
...
30  {'objective': 'binary:logistic', 'base_score
':...
31  {'objective': 'binary:logistic', 'base_score
':...
```

```
32  {'activation': 'elu', 'batch_size': 64, 'dropo
...
Leaderboard.
================ Leaderboard ================
                          Model Name  Metric Score
0          Ensemble: Rank Avg Top 5      0.911357
1              Ensemble: Mean Top 5      0.911291
2   Ensemble: Stacking Top 3 Unique      0.909958
3     Ensemble: Stacking Top 3 Best      0.909958
4             Ensemble: Champ+Rebel      0.908591
5         NeuralNetTorch_tabarena_1      0.799648
6                  xgboost_FLAML_2      0.779852
7                 lightgbm_FLAML_3      0.779246
8              catboost_tabarena_3      0.778021
9              lightgbm_tabarena_1      0.777485
10              xgboost_tabarena_1      0.777365
11                 xgboost_FLAML_1      0.777146
12             catboost_tabarena_2      0.774093
13                lightgbm_FLAML_2      0.773632
14                lightgbm_memento      0.773161
15                    knn_tabarena      0.770553
16          randomforest_tabarena_1      0.769051
17             extraTree_tabarena_3      0.768965
18              log_reg_tabarena_2      0.767381
19              log_reg_tabarena_4      0.767328
20                lightgbm_FLAML_1      0.767116
21              log_reg_tabarena_1      0.767037
22              log_reg_tabarena_3      0.766953
23              log_reg_tabarena_4      0.766866
24                 xgboost_FLAML_3      0.765349
25             catboost_tabarena_1      0.765323
26          randomforest_tabarena_2      0.764499
27          randomforest_tabarena_3      0.761785
28                 catboost_memento      0.745556
29             lightgbm_tabarena_2      0.735472
30              xgboost_tabarena_2      0.734862
31                 xgboost_FLAML_4      0.722846
32        NeuralNetTorch_tabarena_2      0.718918

Process finished with exit code 0
```